System Hardening & Access Control — Firewall Rules

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Platform: Windows Defender Firewall with Advanced Security (Windows host). Test client: Android phone on the same Wi-Fi network. Target: 192.168.1.14.

Objective and configuration

Configure a default-deny inbound firewall policy and permit only the specified lab services: SSH (TCP 22), HTTP (TCP 80), and HTTPS (TCP 443). The Windows Domain, Private, and active Public profiles were confirmed enabled with unmatched inbound connections blocked and unmatched outbound connections allowed. Three enabled inbound allow rules were added for TCP 22, 80 and 443, scoped to LocalSubnet. An explicit inbound block rule for TCP 9999 was created as a negative control.

Rules and observed connection tests

TCP port	Rule	Android test result
22 — SSH	Allow / LocalSubnet	Open (port scanner)
80 — HTTP	Allow / LocalSubnet	Open; Node.js received connections
443 — HTTPS	Allow / LocalSubnet	Open (port scanner)
9999 — control	Block / LocalSubnet	Not open; Chrome timed out

Method and interpretation

Four temporary Node.js TCP listeners were bound to the laptop’s Wi-Fi address (ports 22, 80, 443 and 9999). The Android device scanned these ports from 192.168.1.12. The scanner reported three open and one closed/not-open. Chrome successfully reached TCP 80; the Node.js console logged incoming connections. A request to TCP 9999 timed out and the listener recorded no connection. These observations are consistent with the intended firewall behavior.

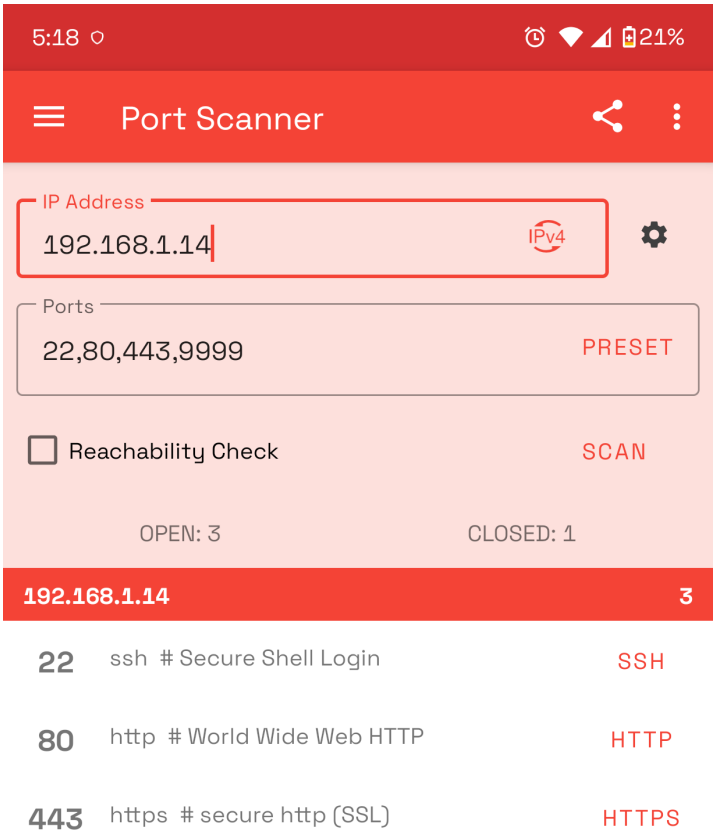
Limitations and conclusion

The listeners tested TCP reachability, not functional SSH authentication or HTTPS encryption. Existing Windows application exceptions were not removed, so the exercise does not establish that all other possible inbound ports are blocked. Within the tested scope, the required ports were reachable and the explicitly denied control port was not. The test listeners and lab-specific rules should be removed after evidence collection.

Evidence: Android PortDroid scan (3 open, 1 not open) attached as Appendix A in the combined submission. Additional supporting evidence: Windows Firewall profile overview, PowerShell rule verification, and Chrome TCP 9999 timeout screenshot (if retained).

Appendix A — Android port-scan evidence

Original screenshot provided by the student. Target: 192.168.1.14; scan ports: 22, 80, 443, 9999. Scanner reports OPEN: 3, CLOSED: 1.



Note: Scanner service names are conventional port labels. Temporary TCP listeners, not production SSH/HTTP/HTTPS services, were used.